(*iv*) The "cosine *i* sine" part of the polar form of a complex number is sometimes abbreviated cis. That is,

$$z = r(\cos\theta + i\sin\theta) = r\,\text{cis}\,\theta.$$

This notation, used mainly in engineering, will not be used in this text.

EXERCISES 1.3

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–10, write the given complex number in polar form first using an argument $\theta \neq \text{Arg}(z)$ and then using $\theta = \text{Arg}(z)$.

1. 2

2. -10

3. $-3i$

4. $6i$

5. $1 + i$

6. $5 - 5i$

7. $-\sqrt{3} + i$

8. $-2 - 2\sqrt{3}i$

9. $\dfrac{3}{-1+i}$

10. $\dfrac{12}{\sqrt{3}+i}$

In Problems 11 and 12, use a calculator to write the given complex number in polar form first using an argument $\theta \neq \text{Arg}(z)$ and then using $\theta = \text{Arg}(z)$.

11. $-\sqrt{2} + \sqrt{7}i$

12. $-12 - 5i$

In Problems 13 and 14, write the complex number whose polar coordinates $(r,\ \theta)$ are given in the form $a + ib$. Use a calculator if necessary.

13. $(4,\ -5\pi/3)$

14. $(2,\ 2)$

In Problems 15–18, write the complex number whose polar form is given in the form $a + ib$. Use a calculator if necessary.

15. $z = 5\left(\cos\dfrac{7\pi}{6} + i\sin\dfrac{7\pi}{6}\right)$

16. $z = 8\sqrt{2}\left(\cos\dfrac{11\pi}{4} + i\sin\dfrac{11\pi}{4}\right)$

17. $z = 6\left(\cos\dfrac{\pi}{8} + i\sin\dfrac{\pi}{8}\right)$

18. $z = 10\left(\cos\dfrac{\pi}{5} + i\sin\dfrac{\pi}{5}\right)$

In Problems 19 and 20, use (6) and (7) to find z_1z_2 and z_1/z_2. Write the number in the form $a + ib$.

19. $z_1 = 2\left(\cos\dfrac{\pi}{8} + i\sin\dfrac{\pi}{8}\right),\ z_2 = 4\left(\cos\dfrac{3\pi}{8} + i\sin\dfrac{3\pi}{8}\right)$

20. $z_1 = \sqrt{2}\left(\cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}\right),\ z_2 = \sqrt{3}\left(\cos\dfrac{\pi}{12} + i\sin\dfrac{\pi}{12}\right)$

In Problems 21–24, write each complex number in polar form. Then use either (6) or (7) to obtain the polar form of the given number. Finally, write the polar form in the form $a + ib$.

21. $(3 - 3i)(5 + 5\sqrt{3}i)$

22. $(4 + 4i)(-1 + i)$

23. $\dfrac{-i}{1+i}$

24. $\dfrac{\sqrt{2} + \sqrt{6}i}{-1 + \sqrt{3}i}$

In Problems 25–30, use (9) to compute the indicated powers. Write the complex number in polar form and then in the form $a + ib$.

25. $(1 + \sqrt{3}i)^9$

26. $(2 - 2i)^5$

27. $\left(\frac{1}{2} + \frac{1}{2}i\right)^{10}$

28. $(-\sqrt{2} + \sqrt{6}i)^4$

29. $\left[\left(\sqrt{2}\cos\dfrac{\pi}{8} + i\sqrt{2}\sin\dfrac{\pi}{8}\right)\right]^{12}$

30. $\left[\sqrt{3}\left(\cos\dfrac{2\pi}{9} + i\sin\dfrac{2\pi}{9}\right)\right]^{6}$

In Problems 31 and 32, write the given complex number in polar form and then in the form $a + ib$.

31. $\left(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9}\right)^{12}\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^{5}$

32. $\dfrac{\left[8\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)\right]^{3}}{\left[2\left(\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right)\right]^{10}}$

33. Use de Moivre's formula (10) with $n = 2$ to find trigonometric identities for $\cos 2\theta$ and $\sin 2\theta$.

34. Use de Moivre's formula (10) with $n = 3$ to find trigonometric identities for $\cos 3\theta$ and $\sin 3\theta$.

In Problems 35 and 36, find a positive integer n for which the equality holds.

35. $\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{n} = -1$ **36.** $\left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)^{n} = 1$

37. For the complex numbers $z_1 = -1$ and $z_2 = 5i$, verify that:

(a) $\text{Arg}(z_1 z_2) \neq \text{Arg}(z_1) + \text{Arg}(z_2)$

(b) $\text{Arg}(-z_2/z_1) \neq \text{Arg}(-z_2) - \text{Arg}(z_1)$.

38. For the complex numbers given in Problem 37, verify that:

(a) $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

(b) $\arg(-z_2/z_1) = \arg(-z_2) - \arg(z_1)$.

Focus on Concepts

39. Suppose that $z = r(\cos\theta + i\sin\theta)$. Describe geometrically the effect of multiplying z by a complex number of the form $z_1 = \cos\alpha + i\sin\alpha$, when $\alpha > 0$ and when $\alpha < 0$.

40. Suppose $z = \cos\theta + i\sin\theta$. If n is an integer, evaluate $z^n + \bar{z}^n$ and $z^n - \bar{z}^n$.

41. Write an equation that relates $\arg(z)$ to $\arg(1/z)$, $z \neq 0$.

42. Are there any special cases in which $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$? Prove your assertions.

43. How are the complex numbers z_1 and z_2 related if $\arg(z_1) = \arg(z_2)$?

44. Describe the set of points z in the complex plane that satisfy $\arg(z) = \pi/4$.

45. Student A states that, even though she can't find it in the text, she thinks that $\arg(\bar{z}) = -\arg(z)$. For example, she says, if $z = 1 + i$, then $\bar{z} = 1 - i$ and $\arg(z) = \pi/4$ and $\arg(\bar{z}) = -\pi/4$. Student B disagrees because he feels that he has a counterexample: If $z = i$, then $\bar{z} = -i$; we can take $\arg(i) = \pi/2$ and $\arg(-i) = 3\pi/2$ and so $\arg(i) \neq -\arg(-i)$. Take sides and defend your position.

46. Suppose z_1, z_2, and $z_1 z_2$ are complex numbers in the first quadrant and that the points $z = 0$, $z = 1$, z_1, z_2, and $z_1 z_2$ are labeled O, A, B, C, and D, respectively. Study the formula in (6) and then discuss how the triangles OAB and OCD are related.

47. Suppose $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 = z_2$, then how are r_1 and r_2 related? How are θ_1 and θ_2 related?

48. Suppose z_1 is in the first quadrant. For each z_2, discuss the quadrant in which $z_1 z_2$ could be located.

(a) $z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ **(b)** $z_2 = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$

(c) $z_2 = -i$ **(d)** $z_2 = -1$

49. (**a**) For $z \neq 1$, verify the identity

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z}.$$

(**b**) Use part (a) and appropriate results from this section to establish that

$$1 + \cos\theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin\left(n + \frac{1}{2}\right)\theta}{2\sin\frac{1}{2}\theta}$$

for $0 < \theta < 2\pi$. The foregoing result is known as **Lagrange's identity** and is useful in the theory of Fourier series.

50. Suppose z_1, z_2, z_3, and z_4 are four distinct complex numbers. Interpret geometrically:

$$\arg\left(\frac{z_1 - z_2}{z_3 - z_4}\right) = \frac{\pi}{2}.$$

51. For n an even positive integer, establish the formula

$$\cos n\theta = \cos^n\theta - \binom{n}{2}\cos^{n-2}\theta\sin^2\theta + \binom{n}{4}\cos^{n-4}\theta\sin^4\theta + \cdots + (-1)^{n/2}\sin^n\theta$$

where $\binom{n}{k}$ is the **binomial coefficient** defined by

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}.$$

Computer Lab Assignments

In Problems 52–55, use a CAS to find the modulus and principal argument of the given complex number. In *Mathematica* use the commands **Abs** and **Arg**.

52. $2 - 2i$

53. $\dfrac{-6i}{1 + \sqrt{3}i}$

54. $3.2 - 7.1i$

55. $\dfrac{(1.4 + 2.0i)(0.3 - 1.2i)}{6.5 - 3.9i}$

1.4 Powers and Roots

INTRODUCTION Recall from algebra that -2 and 2 are said to be *square roots* of the number 4 because $(-2)^2 = 4$ and $(2)^2 = 4$. In other words, the two square roots of 4 are distinct solutions of the equation $w^2 = 4$. In like manner we say $w = 3$ is a cube root of 27 since $w^3 = 3^3 = 27$. This last equation points us again in the direction of complex variables since any real number has only *one real* cube root and *two complex* roots. In general, we say that a number w is an ***n*th root** of a nonzero complex number z if $w^n = z$, where n is a positive integer. For example, you are urged to verify that $w_1 = \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i$ and $w_2 = -\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}i$ are the two square roots of the complex number $z = i$ because $w_1^2 = i$ and $w_2^2 = i$. See Problem 39 in Exercises 1.1.

We will now demonstrate that there are exactly n solutions of the equation $w^n = z$.

Roots

Suppose $z = r(\cos\theta + i\sin\theta)$ and $w = \rho(\cos\phi + i\sin\phi)$ are polar forms of the complex numbers z and w. Then, in view of (9) of Section 1.3, the equation $w^n = z$ becomes

$$\rho^n(\cos n\phi + i\sin n\phi) = r(\cos\theta + i\sin\theta). \tag{1}$$